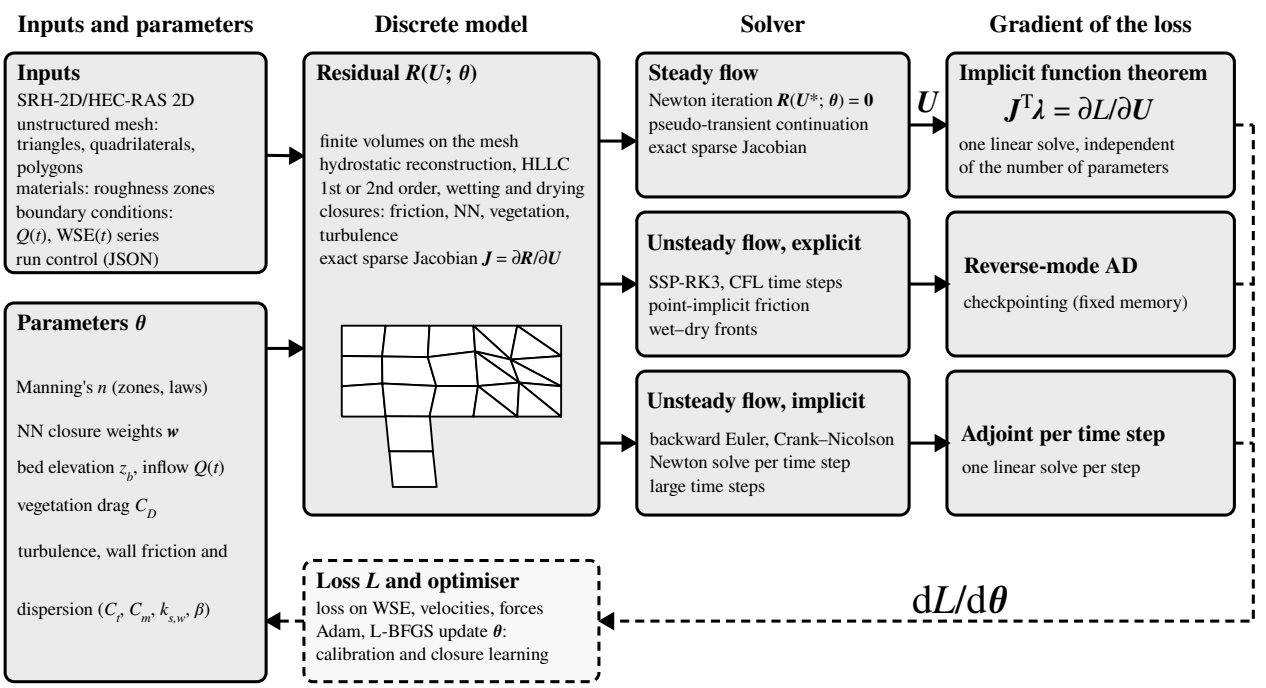
Inputs and parameters
Discrete model
Solver
Gradient of the loss
Inputs
SRH-2D/HEC-RAS 2D
unstructured mesh:
triangles, quadrilaterals,
polygons
materials: roughness zones
boundary conditions:
Q(t), WSE(t) series
run control (JSON)
Parameters θ
Manning's n (zones, laws)
NN closure weights w
bed elevation z_b, inflow Q(t)
vegetation drag C_D
turbulence, wall friction and
dispersion (C_t, C_m, k_s,w, β)
Residual R(U; θ)
finite volumes on the mesh
hydrostatic reconstruction, HLLC
1st or 2nd order, wetting and drying
closures: friction, NN, vegetation,
turbulence
exact sparse Jacobian J = ∂R/∂U
Steady flow
Newton iteration R(U*; θ) = 0
pseudo-transient continuation
exact sparse Jacobian
U
Unsteady flow, explicit
SSP-RK3, CFL time steps
point-implicit friction
wet–dry fronts
Unsteady flow, implicit
backward Euler, Crank–Nicolson
Newton solve per time step
large time steps
Implicit function theorem
J^T λ = ∂L/∂U
one linear solve, independent
of the number of parameters
Reverse-mode AD
checkpointing (fixed memory)
Adjoint per time step
one linear solve per step
dL/dθ
Loss L and optimiser
loss on WSE, velocities, forces
Adam, L-BFGS update θ:
calibration and closure learning